FINITE TYPE ON REDUCED FORMAL SPECTRA IS A LOCAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is complete for all three clauses

Editorial note. The label lemma-finite-type-red names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the WAdm^count category and the ideals of topologically nilpotent elements
- formal spectra and their reduced closed subschemes
- etale charts of affine formal algebraic spaces and etaleness of the induced maps on reductions
- locality on source and target in the etale topology for locally finite type morphisms

Source reference. Original section 0GB6: sources/stacks-chow/tags.txt, source lines 20361–20361. Prerequisite pointer invoked at proof line(s) 2676; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN8: sources/stacks-chow/tags.txt, source lines 13431–13431. Prerequisite pointer invoked at proof line(s) 2663; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AN9: sources/stacks-chow/tags.txt, source lines 13432–13432. Prerequisite pointer invoked at proof line(s) 2677; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GB7: sources/stacks-chow/tags.txt, source lines 20362–20362. Prerequisite pointer invoked at proof line(s) 2674; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CBA: sources/stacks-chow/tags.txt, source lines 15465–15465. Prerequisite pointer invoked at proof line(s) 2639; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 03XE: sources/stacks-chow/tags.txt, source lines 5145–5145. Prerequisite pointer invoked at proof line(s) 2692; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *For an arrow $\varphi : A \to B$ in $WAdm^{count}$ consider the property $P(\varphi) =$"the induced ring homomorphism $A/\mathfrak{a} \to B/\mathfrak{b}$ is of finite type" where $\mathfrak{a} \subset A$ and $\mathfrak{b} \subset B$ are the ideals of topologically nilpotent elements. Then P is a local property as defined in Formal Spaces, Situation 0CBA.*

Proof. Consider a commutative diagram

$$\begin{array}{ccc} B & \longrightarrow & (B')^\wedge \\ {\scriptstyle\varphi}\uparrow & & \uparrow{\scriptstyle\varphi'} \\ A & \longrightarrow & (A')^\wedge \end{array}$$

as in Formal Spaces, Situation 0CBA. Taking Spf of this diagram we obtain

$$\begin{array}{ccc} \mathrm{Spf}(B) & \longleftarrow & \mathrm{Spf}((B')^\wedge) \\ \downarrow & & \downarrow \\ \mathrm{Spf}(A) & \longleftarrow & \mathrm{Spf}((A')^\wedge) \end{array}$$

of affine formal algebraic spaces whose horizontal arrows are representable by algebraic spaces and étale by Formal Spaces, Lemma 0AN8. Hence we obtain a commutative diagram of affine schemes

$$\begin{array}{ccc} \mathrm{Spf}(B)_{red} & \xleftarrow[g]{} & \mathrm{Spf}((B')^\wedge)_{red} \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ \mathrm{Spf}(A)_{red} & \longleftarrow & \mathrm{Spf}((A')^\wedge)_{red} \end{array}$$

whose horizontal arrows are étale by Formal Spaces, Lemma 0GB7. By Formal Spaces, Example 0GB6 and Lemma 0AN9 conditions (1), (2), and (3) of Formal Spaces, Situation 0CBA translate into the following statements

(1) if f is locally of finite type, then f' is locally of finite type,
(2) if f' is locally of finite type and g is surjective, then f is locally of finite type, and
(3) if $T_i \to S$, $i = 1, \ldots, n$ are locally of finite type, then $\coprod_{i=1,\ldots,n} T_i \to S$ is locally of finite type.

Properties (1) and (2) follow from the fact that being locally of finite type is local on the source and target in the étale topology, see discussion in Morphisms of Spaces, Section 03XE. Property (3) is a straightforward consequence of the definition. □